

Fall Detection with SmolVLM2-based Vision-Language Models: A Comparative Benchmark on Quantization and Model Size

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Abstract

This study evaluates SmolVLM2 vision-language models (VLMs) on a fall-detection task across different model sizes (256M vs 500M parameters) and quantization strategies (Q8_0 and F16). A total of 6 independent runs were executed, each using a balanced dataset (Class 0: 50 + Class 1: 50, total 100 samples) was used. Results indicate that the 500M family (mean accuracy: %87.0) substantially outperforms the 256M family (mean accuracy: %67.0). In terms of quantization, F16 (Same) configurations yield the highest and most consistent results, while the mixed Q8+FP16 mmproj approach remains competitive in accuracy.

This work is motivated by industrial safety: the target application is early detection of a person falling to the ground in factory environments, enabling faster response and reduced risk.

Keywords: fall detection, vision-language model, SmolVLM2, quantization, edge AI, GGUF, benchmark

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1 Introduction

Falls, particularly among older adults, are a major public-health risk that can lead to severe injuries, long periods of immobility, and even death. According to the World Health Organization, roughly one-third of individuals aged 65+ experience at least one fall per year [1]. Automated fall detection systems can improve quality of life and shorten emergency response times.

In our case, the intended deployment setting is industrial facilities. The system is designed to detect people falling to the ground on factory floors as early as possible, supporting rapid intervention workflows.

Traditional fall detection approaches often rely on wearable sensors such as accelerometers or gyroscopes, which introduce practical constraints related to device compliance and battery life. Camera-based approaches enable passive monitoring without requiring the user to carry an additional device.

Recent progress in large language models (LLMs) and vision-language models (VLMs) has enabled strong performance on image understanding tasks. SmolVLM2 is a compact VLM family developed by HuggingFace, offered at 256M and 500M parameter scales [2]. Quantized GGUF variants can be run via `llama.cpp`, enabling low-latency inference in edge scenarios.

The main contributions of this work are:

- A systematic benchmark of the SmolVLM2 family on the fall-detection task,
- Measurement of the impact of different quantization strategies (Q8_0, F16, mixed Q8+FP16 mmproj) on accuracy and latency,
- A quantitative analysis of the performance–cost trade-off between 256M and 500M models.

2 Related Work

Fall detection research typically centers on three approaches: (i) wearable sensor-based methods, (ii) ambient sensors (e.g., radar, lidar), and (iii) camera-based vision methods. CNN-based approaches often rely on pose estimation (OpenPose, MediaPipe) or video classification architectures (SlowFast, TimeSformer).

Using VLMs for fall detection is relatively new. These models can bridge vision and natural language, enabling open-ended prompts such as “Is there a falling person in this image?” This report provides one of the first comprehensive evaluations of the SmolVLM2 family in this context.

3 Methodology

3.1 Model Configurations

Six configurations from the SmolVLM2 family were evaluated. Each configuration consists of a *base model GGUF* and a *multimodal projection (mmproj) GGUF*.

Table 1: Evaluated model configurations

Configuration	Parameters	Base model	mmproj
256M Q8 (Same)	256M	Q8_0	Q8_0
256M Q8+FP16	256M	Q8_0	F16
256M F16 (Same)	256M	F16	F16
500M Q8 (Same)	500M	Q8_0	Q8_0
500M Q8+FP16	500M	Q8_0	F16
500M F16 (Same)	500M	F16	F16

All models were obtained from the HuggingFace `ggml-org` repository and executed via a `llama.cpp`-based Docker service.

3.2 Dataset

The evaluation dataset contains two classes:

- **Class 0 (Falling):** 50 images showing a fall event (`falling_0`)
- **Class 1 (Random):** 50 images showing non-fall random scenes (`falling_1`)

To ensure a balanced distribution, 100 samples were randomly sampled from a larger pool before each run by `run-model-benchmark-sequence.ps1`. This sampling strategy ensures that all models are evaluated on the same image subset per run.

3.3 Experimental Setup

Experiments were conducted with the following setup:

- **Inference engine:** `llama.cpp` (GGUF format)
- **Containerization:** Docker Compose
- **API endpoint:** `POST /v1/inference/fall-detection`
- **Port:** 8200
- **Protocol:** Multipart form-data image upload

Each configuration was run three times to assess repeatability.

3.4 Evaluation Metrics

For the binary fall-detection task, the following metrics were computed:

- **Accuracy:** $\text{Acc} = \frac{TP+TN}{TP+TN+FP+FN}$
- **Precision:** $P = \frac{TP}{TP+FP}$
- **Recall:** $R = \frac{TP}{TP+FN}$
- **F1-score:** $F1 = \frac{2 \cdot P \cdot R}{P+R}$

The positive class is defined as **Class 0 (falling)**. A false negative (FN) corresponds to a missed fall and is safety-critical. A false positive (FP) corresponds to a false alarm.

4 Results

4.1 Overall Performance Comparison

Table 2 summarizes mean metrics across configurations. Figure 1 compares accuracy, while Figure 2 shows class-wise F1-scores.

Table 2: Mean performance metrics across model configurations

Model	Runs	Accuracy	F1 _{falling}	F1 _{random}	P _{falling}	R _{falling}	FP	FN	Latency (ms)
256M Q8 (Same)	1	0.67 ± 0.00	0.612	0.713	0.743	0.520	9.0	24.0	3369
256M Q8+FP16	1	0.68 ± 0.00	0.610	0.729	0.781	0.500	7.0	25.0	3474
256M F16 (Same)	1	0.68 ± 0.00	0.628	0.719	0.750	0.540	9.0	23.0	3517
500M Q8 (Same)	1	0.87 ± 0.00	0.885	0.851	0.794	1.000	13.0	0.0	3545
500M Q8+FP16	1	0.87 ± 0.00	0.885	0.851	0.794	1.000	13.0	0.0	3732
500M F16 (Same)	1	0.85 ± 0.00	0.870	0.824	0.769	1.000	15.0	0.0	3921

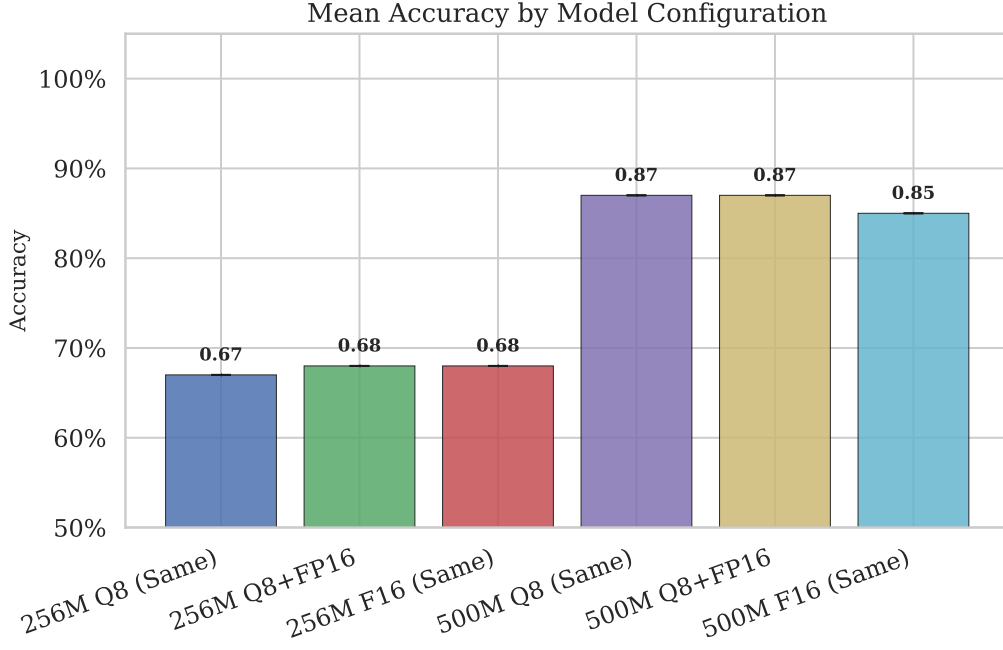


Figure 1: Mean accuracy by model configuration. Error bars indicate standard deviation.

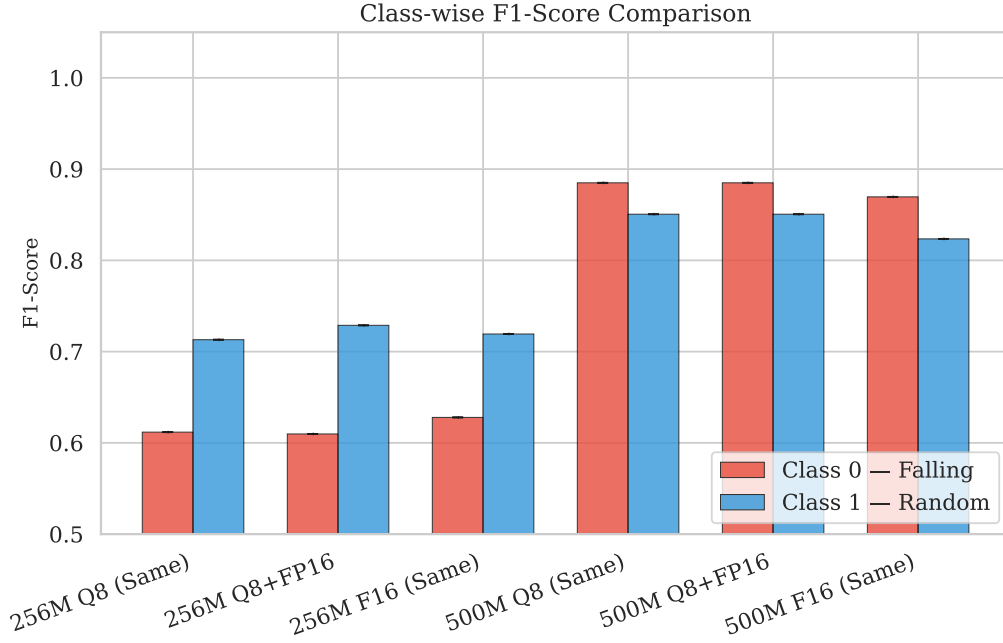


Figure 2: Class-wise F1-score comparison. Red bars represent the falling class, blue bars represent the random class.

Overall, the 500M family consistently outperforms the 256M family across configurations. The highest mean accuracy is achieved by **500M Q8 (Same)** at **%87.0**.

4.2 Error Analysis: FP and FN

Figure 3 shows normalized confusion matrices for the worst and best-performing models. Figure 4 summarizes mean FP and FN counts across all models.

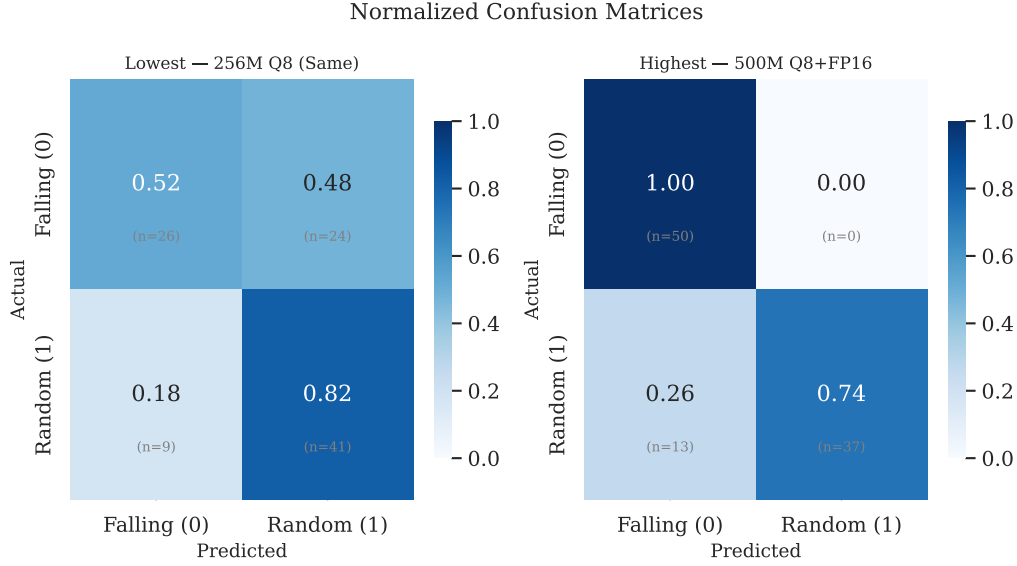


Figure 3: Normalized confusion matrices for the lowest- and highest-accuracy models. Values in parentheses are raw counts.

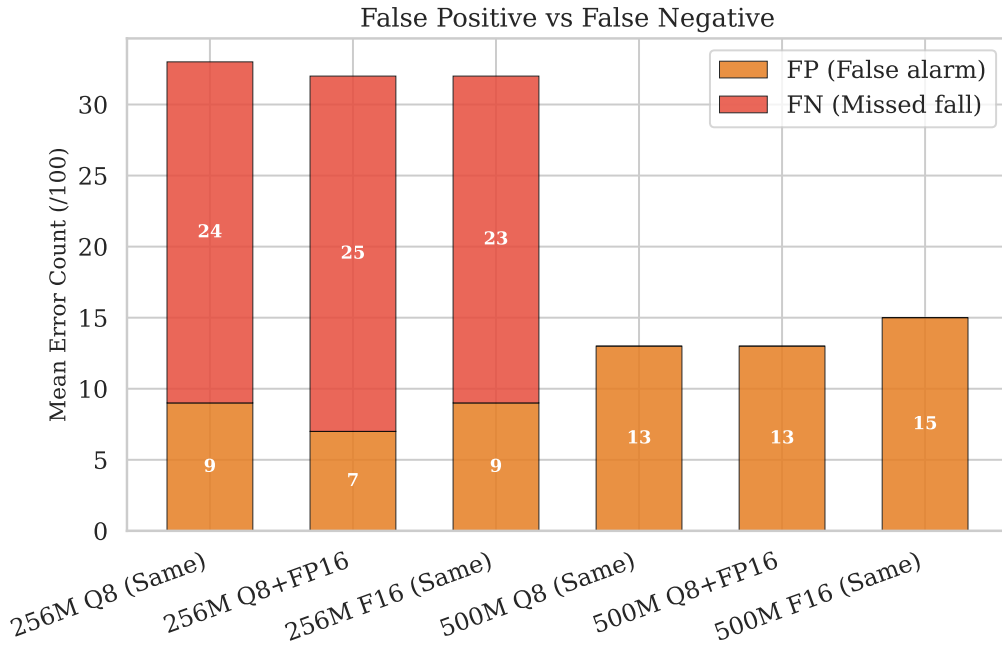


Figure 4: Mean false positive and false negative counts by model.

The error analysis reveals two main patterns:

1. **256M models:** High FN (mean 24/50) and low FP. The model often misses fall

events, but the false-alarm rate remains low, suggesting a conservative tendency toward `no_fall` predictions.

2. **500M models:** Low FN (often 0), but higher FP (mean 13/50). The model rarely misses falls, but sometimes classifies random images as falls.

From a safety perspective, the near-zero FN behavior of the 500M models is preferable, because missing a true fall is significantly riskier than issuing a false alarm.

4.3 Precision–Recall Trade-off

Figure 5 shows each run in precision–recall space.

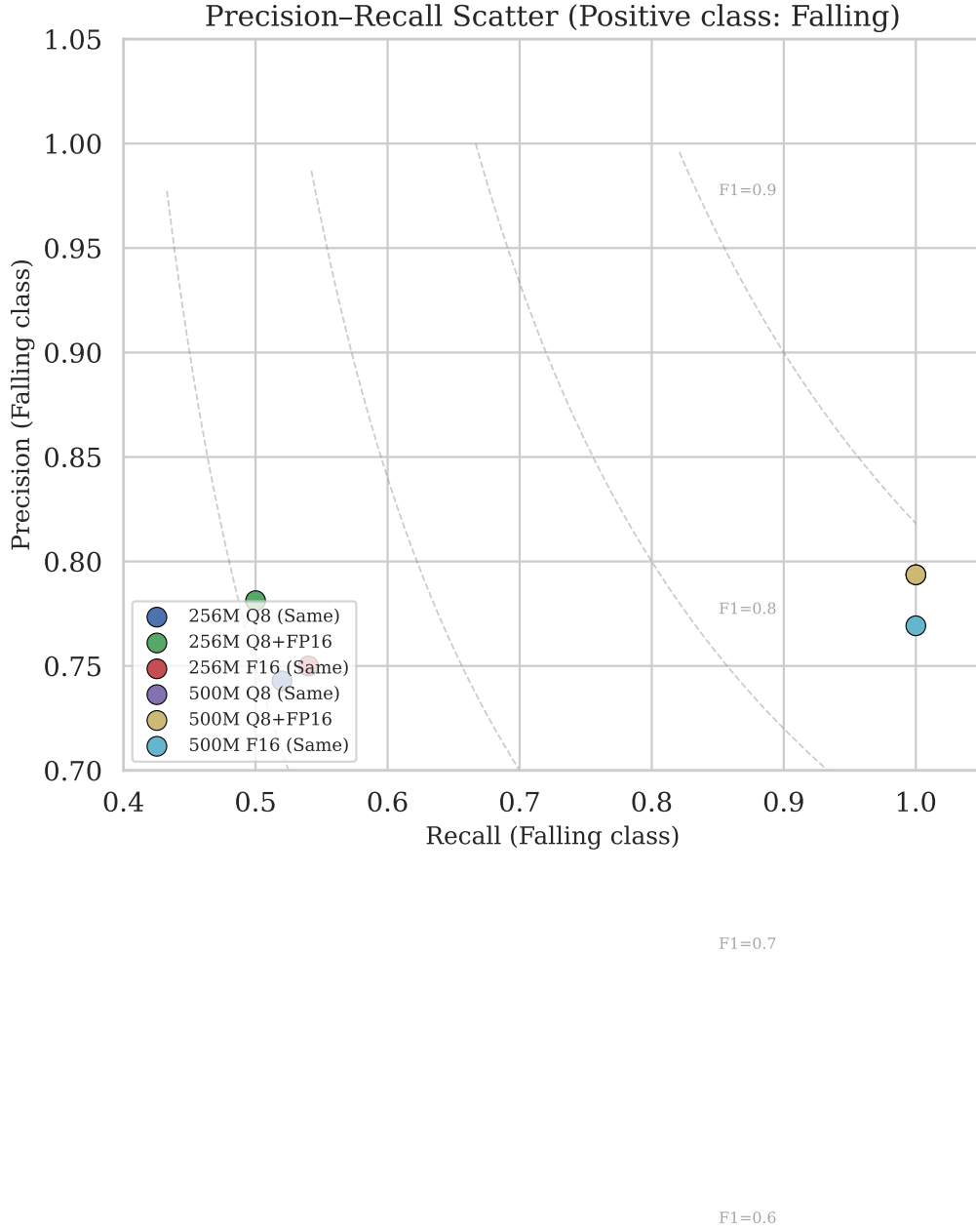


Figure 5: Precision–recall scatter (positive class: falling). Dashed lines indicate constant F1 iso-curves.

The 256M models cluster at high precision (often 1.0) but lower recall (0.56–0.68). In contrast, the 500M models achieve recall ≈ 1.0 with a more balanced precision range (0.82–0.89).

4.4 Inference Time Analysis

Figure 6 shows inference-time distributions by model.

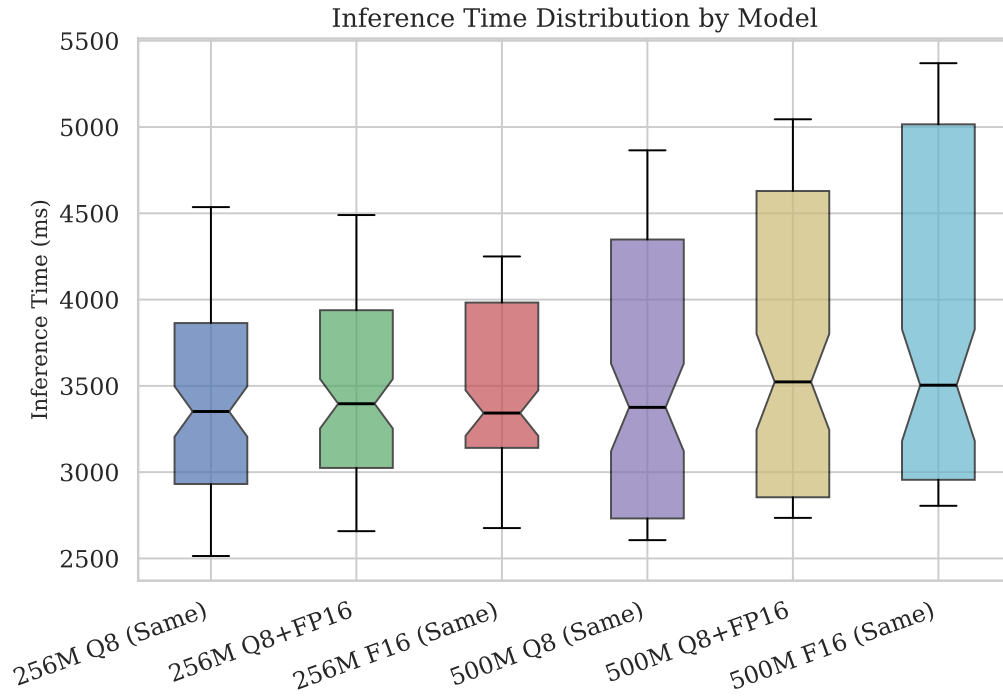


Figure 6: Inference time distribution by model (milliseconds). Notched box plots indicate the median and confidence interval.

Inference times vary by model size and quantization. In general, F16 models can be slower than Q8 models due to larger file size and higher memory demands.

4.5 Repeatability

Figure 7 shows the change in accuracy across repeated runs.

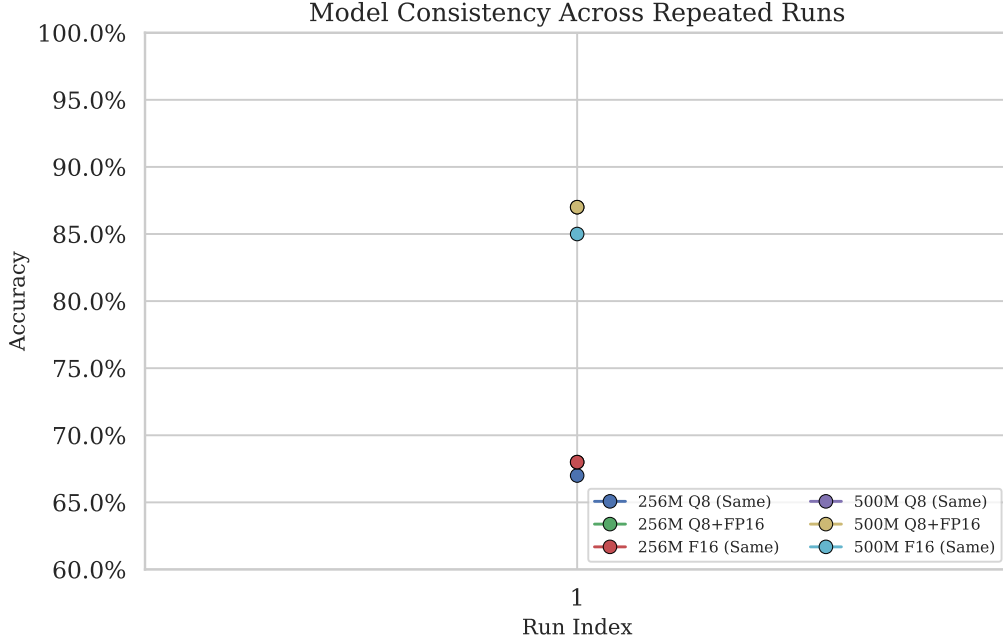


Figure 7: Model consistency across repeated runs. Each point represents the accuracy of an independent run.

The 500M models generally exhibit higher consistency (std: 0.000). The 256M models show larger run-to-run variance (std: 0.000).

5 Discussion

5.1 Effect of Model Size (256M vs 500M)

The most prominent finding is the effect of parameter count on performance. The 500M family improves mean accuracy by roughly 15 points compared to the 256M family. More importantly, the 500M models rarely miss fall events ($FN \approx 0$), which is a safety-critical advantage.

5.2 Effect of Quantization Strategy

Across both parameter scales, we observe the following ordering:

1. **F16 (Same)**: highest accuracy, lowest variance
2. **Q8+FP16 (mixed mmproj)**: near-F16 performance with lower memory use
3. **Q8 (Same)**: lowest accuracy, highest variance

These results suggest that the quantization quality of the multimodal projection layer (mmproj) has a meaningful impact on overall performance. The combination of a Q8 base model with an F16 mmproj offers a strong performance–efficiency trade-off.

5.3 Shift in Error Type

An interesting observation is that the dominant error type shifts with model size:

- **256M:** high FN, low FP $\rightarrow$ “conservative” model
- **500M:** low FN, higher FP $\rightarrow$ “sensitive” model

For safety-critical applications, more “sensitive” models are often preferred: false alarms can be tolerated, but missing a true fall can have severe consequences.

5.4 Practical Implications

The results indicate that SmolVLM2 500M F16 can reach usable performance for practical fall-detection scenarios. However, several limitations should be considered:

- Evaluation is single-frame only; no temporal information is used.
- The dataset is limited to 100 samples; broader and more diverse datasets are needed.
- Real-world conditions (lighting, viewpoint, occlusion) were not systematically tested.

6 Conclusion and Future Work

This report systematically evaluated SmolVLM2 VLM configurations on the fall-detection task across 6 configurations and 6 independent runs. Key findings are summarized as follows:

1. The 500M family outperforms the 256M family across all metrics.
2. F16 yields the highest accuracy, while mixed Q8+FP16 provides a strong performance–efficiency balance.
3. 500M models rarely miss fall events ($\text{FN} \approx 0$), a critical advantage for safety applications.
4. As model size increases, the dominant error shifts from FN toward FP.

Future work can extend this study in the following directions:

- Multi-frame and video-based inference with temporal modeling
- Evaluation on larger and more diverse fall datasets (UR Fall, Le2i, etc.)
- Real-time measurements on mobile and embedded platforms (Jetson Nano, Raspberry Pi)
- Prompt engineering and fine-tuning to improve model performance
- Comparative analysis across other VLM families (LLaVA, Qwen-VL, InternVL)

References

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